SMART AGRICULTURE SYSTEM

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INTRODUCTION

The Smart Agriculture System is an IoT-based project designed to automate and optimize farming. It uses NodeMCU and Arduino with sensors to monitor soil moisture, temperature, and humidity in real time. Based on sensor data, it can automatically control irrigation, helping farmers save water, reduce manual work, and improve crop growth.

PROJECT OBJECTIVES:

- Automate Irrigation Process
- Monitor Environmental Conditions in Real-Time
- Optimize Water Usage
- Reduce Human Effort and Dependency

HARDWARE COMPONENTS:

- NodeMCU – for API data (weather forecast)
- Arduino UNO – controls sensors and pump
- Soil Moisture Sensor – detects soil dryness
- DHT11 Sensor – reads temperature & humidity
- LDR Sensor – measures sunlight
- Servo Motor – moves shade based on sunlight
- Relay Module + Water Pump – for irrigation control
- LCD I2C – displays temperature and moisture

WORKING LOGIC:

NodeMCU fetches weather forecast (OpenWeatherMap API).

If rain chance > 50%, it sends a warning to Arduino.

Arduino checks:

Is the soil dry?

Is temperature high and humidity low?

Is rain predicted?

Based on these, it turns the water pump ON/OFF.

FEATURES:

Automated Irrigation	• Water pump is activated automatically when soil moisture falls below a set threshold
Real-Time Environmental Monitoring	• Continuously measures soil moisture, temperature, and humidity using sensors
Energy Efficient	• Operates only when necessary, reducing power and water consumption
User-Defined Threshold	• Easily configurable moisture levels for different types of crops and soil conditions

Arduino UNO code:

ardu.ino

```
#include <DHT.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <Servo.h>
#include <SoftwareSerial.h>

#define SOIL_PIN A2
#define DHTPIN 4
#define DHTTYPE DHT11
#define RELAY_PIN 7
#define LDR_PIN A0
#define SERVO_PIN 9

DHT dht(DHTPIN, DHTTYPE);
LiquidCrystal_I2C lcd(0x27, 16, 2);
Servo shadeServo;
SoftwareSerial nodeSerial(10, 11); // RX, TX for communication with NodeMCU

String rainIn6Hr = "No";
unsigned long previousMillis = 0;
const long interval = 5000; // 5 seconds interval
bool relayActive = false;
unsigned long relayStartMillis = 0;
const unsigned long relayOnDuration = 5000; // Relay ON for 5 seconds

void setup() {
  Serial.begin(9600);
  nodeSerial.begin(9600);

  pinMode(RELAY_PIN, OUTPUT);
  digitalWrite(RELAY_PIN, HIGH); // Keep relay OFF initially (active LOW)
  dht.begin();
```

Output Serial Monitor ×

Message (Enter to send message to 'Arduino Uno' on 'COM8')

```
15:24:21.236 -> Received from NodeMCU: RainIn6Hr:No
15:25:36.772 -> Received from NodeMCU: ◆
15:25:52.562 -> Received from NodeMCU: RainIn6Hr:No
```

ardu.ino

```
  pinMode(RELAY_PIN, OUTPUT);
  digitalWrite(RELAY_PIN, HIGH); // Keep relay OFF initially (active LOW)
  dht.begin();
  lcd.init();
  lcd.backlight();
  shadeServo.attach(SERVO_PIN);

  Serial.println("Smart Farming System Initialized");
}

void loop() {
  // Weather data receive FROM NODEMCU
  if (nodeSerial.available()) {
    String msg = nodeSerial.readStringUntil('\n');
    msg.trim();
    Serial.println("Received from NodeMCU: " + msg);
    if (msg.startsWith("RainIn6Hr")) {
      rainIn6Hr = msg.substring(10);
    }
  }

  // Current time
  unsigned long currentMillis = millis();

  // Non-blocking update every 5 seconds
  if (currentMillis - previousMillis >= interval) {
    previousMillis = currentMillis;

    int soilValue = analogRead(SOIL_PIN);
    int lightValue = analogRead(LDR_PIN);
    float temp = dht.readTemperature();
    float hum = dht.readHumidity();

    lcd.clear();
    lcd.setCursor(0, 0);
    if (soilValue > 900) {
```

Output Serial Monitor ×

Message (Enter to send message to 'Arduino Uno' on 'COM8')

```
15:31:51.255 -> Received from NodeMCU: ◆
15:32:05.438 -> Received from NodeMCU: RainIn6Hr:No
```

ardu.ino

```
    float temp = dht.readTemperature();
    float hum = dht.readHumidity();

    lcd.clear();
    lcd.setCursor(0, 0);
    if (soilValue > 900) {
      lcd.print("Soil: Dry");
    } else {
      lcd.print("Soil: Wet");
    }

    lcd.setCursor(0, 1);
    lcd.print("T:" + String(temp) + " H:" + String(hum));

    // Shade control based on light intensity
    if (lightValue < 200) {
      shadeServo.write(90); // open Shade
    } else {
      shadeServo.write(0);  // close Shade
    }

    // Relay control logic — ON only when condition is true
    if (!relayActive && rainIn6Hr == "No" && soilValue > 900 && hum < 70 && temp > 25) {
      digitalWrite(RELAY_PIN, LOW); // Turn ON relay (active LOW)
      relayStartMillis = currentMillis;
      relayActive = true;
    }
  }

  // Turn OFF relay after 5 seconds
  if (relayActive && (millis() - relayStartMillis >= relayOnDuration)) {
    digitalWrite(RELAY_PIN, HIGH); // Turn OFF relay
    relayActive = false;
  }
}
```

Output Serial Monitor ×

Message (Enter to send message to 'Arduino Uno' on 'COM8')

```
15:31:51.255 -> Received from NodeMCU: ◆
15:32:05.438 -> Received from NodeMCU: RainIn6Hr:No
```

NodeMCU code:

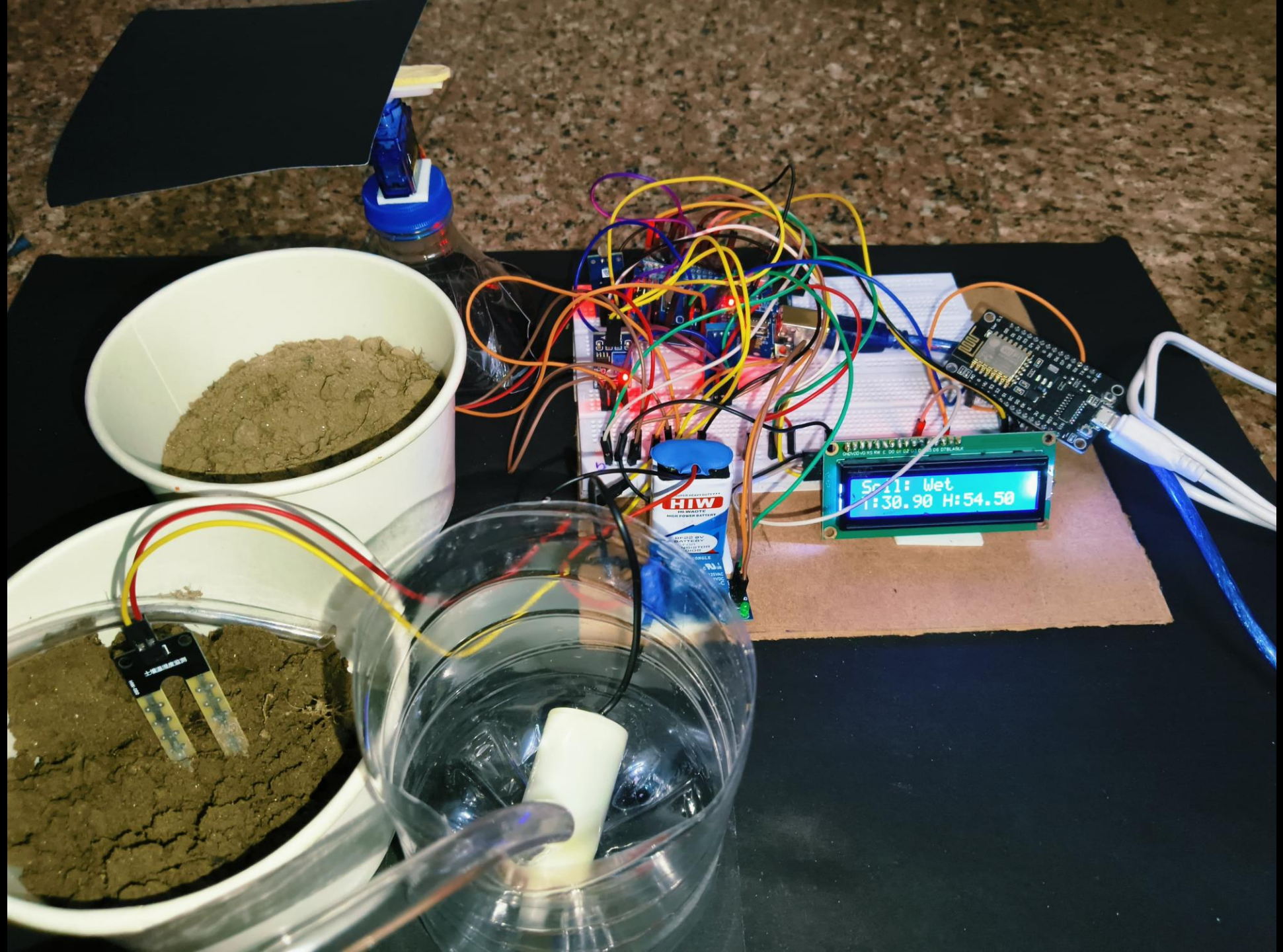
nodemcu.ino

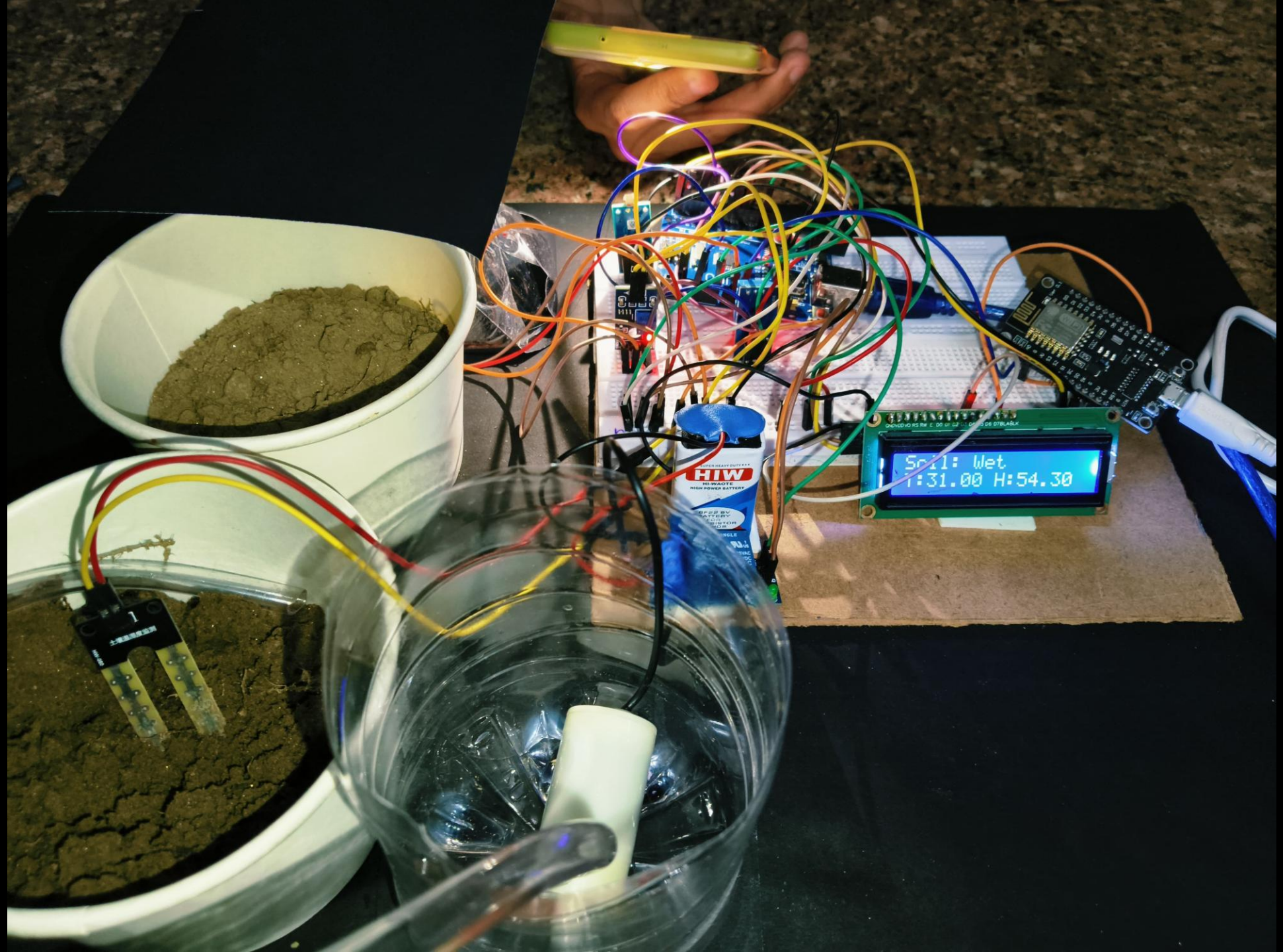
```
#include <ESP8266WiFi.h>
#include <WiFiClient.h>
#include <ArduinoJson.h>
#include <SoftwareSerial.h>

const char* ssid = "Myvivo";
const char* password = "rks12124";
const char* host = "api.openweathermap.org";
const String city = "Delhi";
const String apiKey = "6094ddc5e14b9095281fc6156b745862";

SoftwareSerial mySerial(-1, 5);  // RX, TX for communication with Arduino

void setup() {
  Serial.begin(9600);     // Debugging on USB serial
  mySerial.begin(9600);   // To Arduino

  WiFi.begin(ssid, password);
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
  }

  delay(1000);
  fetchRainNext6Hr(); // Check for rain in next 6 hours
}

void loop() {

}

// Check for rain in next 6 hours
void fetchRainNext6Hr() {
  WiFiClient client;
  if (!client.connect(host, 80)) return;

  String url = "/data/2.5/forecast?q=" + city + "&appid=" + apiKey;
  client.print(String("GET ") + url + " HTTP/1.1\r\nHost: " + host + "\r\nConnection: close\r\n\r\n");

  while (!client.available()) delay(1);
  String payload = client.readString();

```

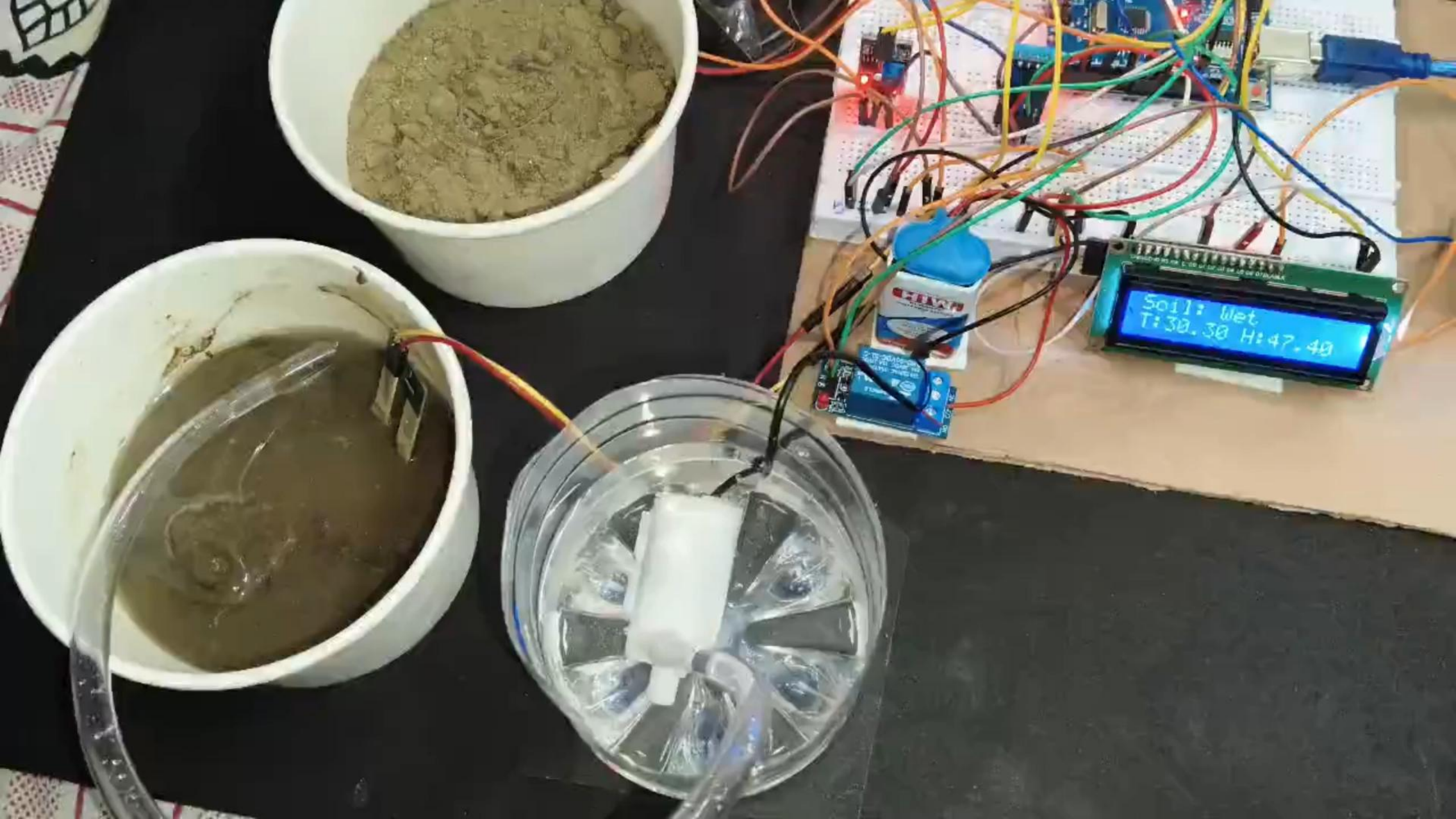
```
void loop() {

}

// Check for rain in next 6 hours
void fetchRainNext6Hr() {
  WiFiClient client;
  if (!client.connect(host, 80)) return;

  String url = "/data/2.5/forecast?q=" + city + "&appid=" + apiKey;
  client.print(String("GET ") + url + " HTTP/1.1\r\nHost: " + host + "\r\nConnection: close\r\n\r\n");

  while (!client.available()) delay(1);
  String payload = client.readString();

  int jsonStart = payload.indexOf('{');
  if (jsonStart > 0) payload = payload.substring(jsonStart);

  StaticJsonDocument<2048> doc;
  if (deserializeJson(doc, payload)) return;

  bool rainExpected = false;
  for (int i = 0; i <= 2; i++) {  // Check the next 6 hours
    const char* cond = doc["list"][i]["weather"][0]["main"];
    if (String(cond).indexOf("Rain") >= 0) {
      rainExpected = true;
      break;
    }
  }

  if (rainExpected)
    mySerial.println("RainIn6Hr:Yes");
  else
    mySerial.println("RainIn6Hr:No");
}
```

WORKING MODEL IMAGES:

Soil: Dry
T:30.90 H:54.80


Soil: Wet
T:30.90 H:54.50
HIW


Soil: Wet
T:31.00 H:54.30
HIW
HI-WAOTE
HIGH POWER BATTERY


Soil: Wet
T:30.30 H:47.40

BENEFITS:

WATER CONSERVATION

SAVES FARMERS' TIME

PROTECTS CROPS FROM HARSH WEATHER

SMART USE OF IOT AND AUTOMATION

PROMOTES SUSTAINABLE FARMING

CONCLUSION

The Smart Agriculture System uses IoT to automate irrigation and monitor farming conditions in real-time. It helps save water, reduce manual effort, and improve crop health. This low-cost, scalable solution promotes smarter and more sustainable farming practices.

THANK YOU